

AMMAR BADLAWALA

Aspiring Security Engineer • SOC / Detection • Networking

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[LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

I ship working security platforms, then break them on purpose to make them stronger — seven so far, including a SIEM, a honeypot, and a machine-learning network anomaly detector. I direct AI-assisted adversarial testing against my own systems and personally verify every fix, closing real vulnerabilities before anyone else finds them. B.Sc. Information Technology graduate seeking an entry-level role in security engineering, SOC/detection, or networking.

TECHNICAL SKILLS

Core: Python, FastAPI, PostgreSQL, React, Git/GitHub, REST API design, JWT auth patterns, Linux fundamentals

Also working with: Nmap, Wireshark, Next.js/TypeScript, MongoDB, scikit-learn, Scapy/PyShark, Docker

FEATURED PROJECTS

Mini SIEM — flagship project — [Live demo](#)

FastAPI • PostgreSQL (Neon) • React 18 + Tailwind + shadcn/ui • own JWT + bcrypt + RBAC

- Built a SIEM from scratch: 11 auto-detected log formats, 15 MITRE ATT&CK-tagged detection rules, and an incident correlation engine that merges an attacker's separate alerts into one case. 600+ backend tests run in CI on every commit.
- Shipped a dev-gated “attack lab” that fires real attacks (SQL injection, XSS, path traversal, brute force) at the running system, so detection is proven against live traffic, not trusted on faith.
- Directed AI-assisted adversarial testing, then personally reviewed and verified the result: 3 of 4 URL-encoded attacks were bypassing signature rules undetected. Fixed and retested.
- Directed a second AI-assisted review post-deployment that caught a genuine IP-spoofing vulnerability in the rate limiter — forged headers could bypass it entirely. Root-caused and patched.

NetSentinel — Network Anomaly Detection — [Live demo](#)

FastAPI • scikit-learn • Scapy / PyShark • React 19 + Tailwind • Supabase

- Built a network-traffic anomaly detector trained on real captured packets, not a synthetic dataset — uploads a PCAP, engineers features, and scores anomalies with a trained ML model (Isolation Forest / One-Class SVM).
- Added a RAG-grounded AI analyst (ChromaDB + LangGraph + Groq) that explains why traffic was flagged and what to do next, instead of returning a raw anomaly score an analyst has to interpret alone.
- Directed AI-assisted testing that uncovered a real audit-log bug — attacker IPs were misattributed due to a proxy-chain misconfiguration. Root-caused and fixed.

ThreatHunter Intelligence Platform — [Live demo](#)

Next.js (App Router, TS) • Express + TypeScript • Supabase (Postgres, Auth, RLS) • Zod

- Built an IOC lookup platform fanning out IP, domain, hash, and URL searches across five real threat-intelligence providers (AbuseIPDB, AlienVault OTX, IPInfo, VirusTotal, NIST NVD), with an AI assistant over investigation history.
- Directed a full AI-assisted adversarial security-testing pass that surfaced 8 real findings; personally reviewed, fixed, and retested every one. Backed by 89 automated tests. Repository is private — available to walk through on request.

HoneyShield Intelligence Platform — HTTP trap live in production

Python asyncio (raw sockets, no framework) • SQLite • Streamlit + Plotly • Gemini

- Built a multi-protocol honeypot (SSH, FTP, HTTP, Telnet) from raw sockets, with a correlation engine that spots coordinated attack campaigns by ASN and time window — not just single-IP brute force.

ADDITIONAL PROJECTS

- **phishguard-ai** — Hybrid phishing-URL detector (Random Forest model + 16-rule heuristic engine) with live threat-intel checks and a working Chrome extension. [GitHub](#)
- **FileShield** — File-intelligence platform (hashing, entropy analysis, YARA matching). First shipped project, deployed live. [Live demo](#) • [GitHub](#)
- **Intelligent Log Analyzer** — Parses SSH/Apache/Windows Event logs and scores threats with a weighted model. Mid-rebuild toward a multi-tenant enterprise version (in progress). [GitHub](#)

CERTIFICATIONS & HANDS-ON PRACTICE

Cisco: Endpoint Security • Network Defense • Networking Basics • Introduction to Cybersecurity

Coursera: AI For Everyone • Launching into Machine Learning • Introduction to AI and Machine Learning on Google Cloud • Getting Started with Linux Terminal • Foundations of Cybersecurity • Introduction to Networking and Cloud Computing

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EDUCATION

Bachelor of Science, Information Technology

Somaiya School of Basic Science & Applied Mathematics (Somaiya Vidyavihar University), Mumbai

2023 – 2026 • CGPA: 9.84 / 10 (A+)